

Analysis First-Year Exam, January 2014, Part 1

Do only 4 problems of the 6. Work must be presented in a neat and logical fashion to receive credit. Do not leave any gaps. State clearly any theorems used in proofs.

1. Let F be a closed subset of the reals $\mathbb{R}$ with the usual topology. Show F is a countable union of compact sets.
2. Let E be a nonempty subset of a metric space. State the definition of the distance $\rho_E(x)$ of a point x to E . Characterize $\overline{E}$ in terms of ρ_E , where $\overline{E}$ is the closure of E .
3. This problem has two parts.
 - (a) Let f be a continuous function from a compact metric space X into a metric space Y . Prove $f(X)$ is compact.
 - (b) Assume the setting in (a) with the further assumption that Y is the reals with the usual topology. Show that f assumes maximum and minimum values on X .
4. Let f be a continuous map of a metric space X into a metric space Y .
 - (a) Show that if f is NOT uniformly continuous on X , then for some $\varepsilon > 0$ there are sequences $\{p_n\}, \{q_n\}$ in X such that $d_Y(f(p_n), f(q_n)) > \varepsilon$ for each n but $d_X(p_n, q_n) \rightarrow 0$.
 - (b) Assume that X is compact. Show, using (a), that f is uniformly continuous on X .
5. **Warning: Suspected error.** Part (b) is not valid if the sequence $\{x_n\}$ contains repeated values: the jump at x_n would include every c_k for which $x_k = x_n$. The intended repair is not unique; the source reading has therefore been retained.

Let $\{x_n\}$ be a sequence of points in (a, b) and let $\{c_n\}$ be a sequence of positive numbers such that $\sum c_n$ converges. Define

$$f(x) = \sum_{n: x_n < x} c_n, \quad a < x < b,$$

where the summation is understood as follows: sum over those indices n for which $x_n < x$. If there are no points $x_n < x$, define the sum to be zero.

- (a) Show $f(x-) = f(x)$ for each x in (a, b) .
 - (b) Show $f(x_n+) - f(x_n-) = c_n$, for each n .
6. Suppose f is continuous on $[0, \infty)$, differentiable on $(0, \infty)$, $f(0) = 0$ and f' is monotonically increasing. Define g on $(0, \infty)$ by $g(x) = f(x)/x$, $x > 0$. Prove g is monotonically increasing.